May 13, 2026

Editors
Journal of Public Economics

Dear Professors Hendren and Kopczuk,

I am submitting "Quantifying the Annuity Puzzle: A Unified Lifecycle Decomposition" for consideration at the *Journal of Public Economics*.

Yaari (1965) proved that rational consumers should fully annuitize under actuarially fair pricing, yet voluntary ownership among US retirees is only a few percent. Six decades of research have proposed partial explanations, but the literature has lacked a unified quantitative accounting of how the main channels interact in a single model. Pashchenko (2013, *JPubE*) came closest, but still predicted participation of roughly 20%.

The paper develops two complementary models. **Model 1** is a structural lifecycle decomposition. A nine-channel rational-plus-preference specification (Social Security, bequests, medical expenditure risk with health-mortality correlation, survival pessimism, pricing loads, inflation erosion, long-term care risk, age-varying consumption needs from Aguiar–Hurst 2013, and state-dependent utility from Finkelstein–Luttmer 2013) predicts ownership of 6.8%. This is an underestimate of the two HRS empirical comparators reported in parallel—3.11% (95% CI $[2.54\%, 3.81\%]$) for the cleaner fat-file lifetime contract indicator and 5.21% (95% CI $[4.45\%, 6.09\%]$) for the conventional any-annuity income proxy used in prior literature, both computed on the same wealth-restricted sample as the model predictions. We do not claim that the model "matches" or "resolves" the puzzle; it accounts for most of the gap through standard mechanisms, with a small residual.

Model 1 extended. Layering two behavioral channels onto the nine-channel baseline yields an eleven-channel specification with predicted ownership of 0.0%. The behavioral channels are source-dependent utility (Blanchett–Finke 2024, 2025) and a purchase-event disutility consistent with the Brown (2008) / Chalmers–Reuter (2012) / Hu–Scott (2007) narrow-framing literature. Both are calibrated to literature magnitudes rather than estimated from US moments; we are explicit that the eleven-channel specification is exploratory and that the structural-baseline-plus-residual reading is the disciplined one.

Model 2 is a reduced-form transport using the UK 2015 pension-freedoms reform as out-of-sample variation. Pre-reform mandatory annuitization sustained DC-pot annuity retention of 95%; post-reform retention is 17% at the mid anchor (range 13%–25% across ABI and FCA industry estimates, supplemented by individual-level ELSA pension-grid evidence). Applied as a behavioral wedge against the frictionless population benchmark of 41.8%, this yields a predicted US ownership of 7.5% (bracket $[5.7\%, 11.0\%]$). The two models reach the same neighborhood from independent directions: structural decomposition from inside the lifecycle problem, and cross-country transport from outside it. The conventional HRS income proxy (5.21%, CI $[4.45\%, 6.09\%]$) sits just below the Model 2 bracket's lower edge, with its upper confidence bound extending into the bracket; the cleaner lifetime-contract indicator falls below the bracket.

The paper makes four contributions. First, the unified structural framework—the first to nest age-varying consumption needs alongside the established rational and preference channels of the annuity-puzzle literature. Second, the cross-validation of structural decomposition against an external reduced-form transport based on the UK 2015 reform, drawing on individual-level ELSA pension-grid records alongside industry aggregates. Third, an exact Shapley decomposition over all $2^9 = 512$ subsets of the nine structural channels (the disciplined headline attribution), with an exploratory eleven-channel extension over $2^{11} = 2{,}048$ subsets that adds the two literature-magnitude behavioral parameters; the behavioral channels enter with large but offsetting contributions (42.8 pp for purchase-event disutility, -30.3 pp for source-dependent utility) and are reported as illustrative rather than identified. Fourth, two distinct policy levers emerge: a supply-side reform (group pricing) raises predicted ownership relative to the structural baseline, and a demand-side reform anchored externally by Model 2 operates on a separate margin; neither is a clean structural identification of the underlying mechanism.

The paper is a good fit for the *Journal of Public Economics*. It engages the journal's core interests in Social Security, retirement saving, insurance-market frictions, and the design of public and employer-sponsored retirement institutions, and it engages directly with Pashchenko (2013) and related work published in *JPubE* and neighboring outlets.

Replication code in Julia and all calibration data are publicly available at `https://github.com/DerekTharp/annuity-puzzle-model`. The manuscript has not been submitted elsewhere.

Sincerely,

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